

Real-Time IK-Based Full-Body and Facial Capture for Stylized 2D Puppets in XR Applications

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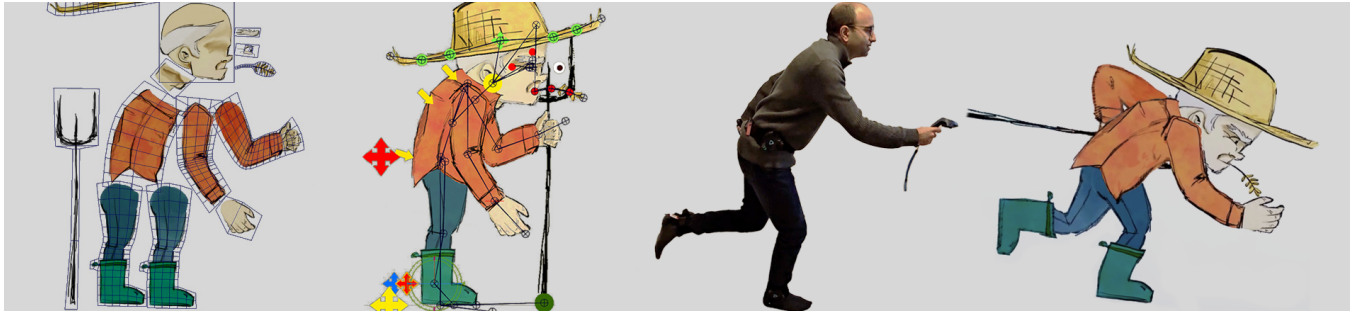


Figure 1: Building 2D puppets for motion capture. Character design ©Deanna Moorehead, used with permission.

Abstract

This demonstration presents the process of setting up 2D characters within game engines, leveraging real-time capabilities to marry traditional 2D and immersive workflows. While earlier efforts have used 3D tools to rig 2D characters, they were limited to linear animation pipelines. This project shows how real-time interactivity can be applied to 2D pipelines. It also introduces two novel features that make the setup adaptable for immersive, hybrid, and interactive applications.

In this demo, visitors will interact with a stylized 2D character using a consumer-grade VR tracking system, experiencing real-time full-body and facial capture. They can recalibrate the character and fine-tune its responsiveness in real time, allowing them to experience how the setup adapts to different body types and movement styles. This hands-on experience showcases how accessible hardware can democratize expressive character control for immersive environments.

CCS Concepts

• **Computing methodologies** → **Motion capture**; *Procedural animation*; **Mixed / augmented reality**; **Virtual reality**; • **Human-centered computing** → **Mixed / augmented reality**; **Virtual reality**; *Gestural input*.

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Keywords

2D puppeting, IK-Based 2D Motion Capture, VR, AR, XR, VP

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1 Introduction

Real-time motion capture pipelines typically operate within 3D environments. By contrast, even with recent advancements in automatically animating 2D figures from pre-recorded motion capture data [Smith et al. 2023], interactive animation and real-time motion capture for 2D puppets remain largely unexplored. Even the few existing real-time 2D puppet workflows are designed for linear applications, or require advanced equipment and extensive parameter adjustments, limiting their adaptability to XR contexts. This demonstration introduces a workflow that merges elements of both worlds, enabling expressive 2D puppetry in real time.

2 Methodology

This demonstration highlights our workflow and the novel techniques we introduced in our system. The methodology is organized into several components, each addressing a specific aspect of our real-time 2D puppetry technique.

2.1 2D Rigging in Game Engine

Traditional 2D puppeting workflows often rely on software with limited rigging and interactivity capabilities. Leveraging 3D pipelines enables more sophisticated rigs, while game engines add real-time responsiveness. This demonstration showcases an animation-ready 2D puppet fully rigged in the game engine, highlighting real-time

interaction via game controllers and VR devices. The result is a reactive, user-driven performance that integrates seamlessly into real-time and non-real-time pipelines.

2.2 Parametric and Dynamic Materials for Drawing Replacements

A critical aspect of 2D puppeting is the ability to switch between drawings to represent facial expressions and gestures. In our workflow, this functionality is implemented with dynamic materials that enable drawing replacements directly within the game engine. Each material holds multiple textures with associated indices, which can be driven by exposed parameters. These parameters may be connected to real-time inputs such as game controllers or facial capture systems (Figure 2). By borrowing the concept of 3D morph targets, these materials respond to live performances, allowing switching between drawings based on inputs. This integration allows 2D puppets to react in real time, supporting expressive performances, which brings the responsiveness and emotional range to the 2D interactive pipeline.

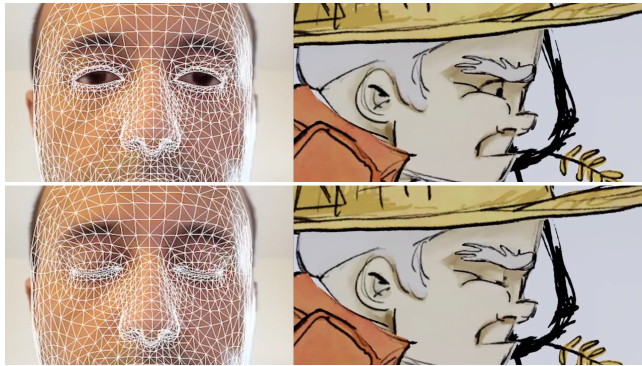


Figure 2: Swapping pre-made drawing replacements by reading real eye blink using facial capture input device.

2.3 Increased Stability and Reduced Complexity Using Novel IK-Based Motion Capture

Traditional motion capture systems often rely on Forward Kinematics, requiring numerous tracking points and resulting in complex, noise-prone setups. Our approach uses Inverse Kinematics with only five trackers, one for each limb and one for the center of gravity, greatly reducing setup complexity. Our 2D framework limits the available axes of rotation. This avoids the need for extra trackers to resolve rotational ambiguities like limb twist, a known complexity in 3D capture systems [Moeslund et al. 2006], thereby minimizing jitter, enhancing stability, and ensuring cleaner performance. Additionally, since the IK limbs operate independently, local tracking errors remain isolated. This prevents the chain reactions of error propagation common in sequential, hierarchical FK systems. [Ignacy Dulęba 2013]. This simplified and robust setup is not only ideal for XR applications where users often rely on consumer-grade trackers, but also enables expressive full-body motion capture with minimal hardware, broadening access to real-time animation workflows, democratizing the motion capture process across linear and real-time applications. (Figure 3)

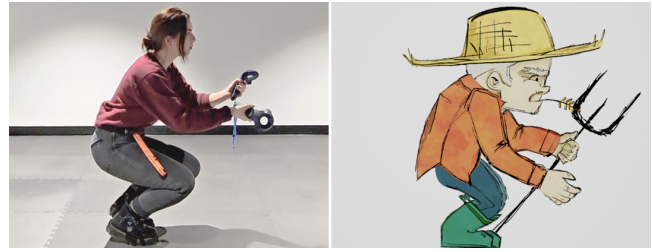


Figure 3: 2D motion capture in action, showing the stability and range of movements in the IK-based MoCap workflow.

2.4 Retargeting, Versatility, and Adaptability for Non-Traditional Characters

Non-traditional characters often present unique challenges in motion capture, particularly when their default poses or degrees of freedom deviate from standard humanoid anatomy. We introduce a “sensitivity” attribute that enables real-time adjustment of captured motion data to accommodate varying character proportions and movement ranges, improving motion fidelity across different designs. Additionally, we introduce a reset-default-pose feature that allows on-the-fly recalibration and retargeting of performances onto characters with unconventional skeletal structures.

Together, these features make the system highly adaptable, supporting a wide range of character designs and performers. Versatility, a key goal in many rigging and motion capture systems, particularly in game applications [Bereznyak 2016], is especially valuable in XR contexts where real-time recalibration and adaptation is important.

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A Supplementary Material

Additional animated examples are available online. While this demonstration focuses on real-time applications, the IK-based approach also simplifies post-capture cleanup, making it well suited for non-real-time pipelines. A cleanup workflow is also provided.

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